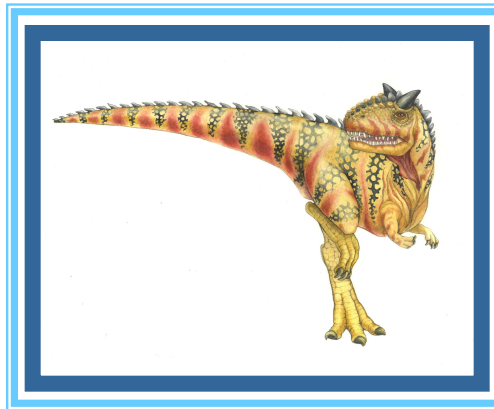


Chapter 8: Main Memory





Chapter 8: Memory Management

- Background
- Swapping
- Contiguous Memory Allocation
- Segmentation
- Paging
- Structure of the Page Table
- Example: The Intel 32 and 64-bit Architectures
- Example: ARM Architecture





Objectives

- To provide a detailed description of various ways of organizing memory hardware
- To discuss various memory-management techniques, including paging and segmentation
- To provide a detailed description of the Intel Pentium, which supports both pure segmentation and segmentation with paging





Dynamic Storage-Allocation Problem

How to satisfy a request of size n from a list of free holes?

- **First-fit**: Allocate the *first* hole that is big enough
- **Best-fit**: Allocate the *smallest* hole that is big enough; must search entire list, unless ordered by size
 - Produces the smallest leftover hole
- **Worst-fit**: Allocate the *largest* hole; must also search entire list
 - Produces the largest leftover hole

First-fit and best-fit better than worst-fit in terms of speed and storage utilization





Placement Algorithm

Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?





Placement Algorithm

First-fit:

212K is put in 500K partition

417K is put in 600K partition

112K is put in 288K partition (new partition $288K = 500K - 212K$)

426K must wait





Placement Algorithm

Best-fit:

212K is put in 300K partition

417K is put in 500K partition

112K is put in 200K partition

426K is put in 600K partition





Placement Algorithm

Worst-fit:

212K is put in 600K partition

417K is put in 500K partition

112K is put in 388K partition

426K must wait

In this example, best-fit turns out to be the best.

